Ian Lingo

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Education

Texas State University San Marcos, TX
B.S. in Computer Science *Graduated: May 2026*

- **GPA**: 3.58
- **Minors**: Applied Mathematics, Mass Communication
- **Honors**: Cum Laude, IEEE-HKN (Eta Kappa Nu), Texas State Honors College
- **Coursework**: Data Structures & Algorithms, Discrete Mathematics, Object-Oriented Programming, Software Engineering, Database Systems, Operating Systems, Systems Security, Computer Architecture

San Jose State University - Open University San Jose, CA
Advanced Python Programming *Jun. 2026 – Aug. 2026*

Technical Skills

Programming Languages: Python, Java, C++, TypeScript, JavaScript, SQL
Frameworks/Tools: Spring Boot, React, Next.js, Flask, Gradle, REST APIs, Tailwind CSS
Testing: JUnit 5, Mockito, Testcontainers, pytest
Cloud/DevOps: Git, GitHub, GitHub Actions, CI/CD, Docker, AWS, OCI, Jira, Vercel, Cloudflare
Data: PostgreSQL, SQL Server, BigQuery, Tableau, Pandas, NumPy, Matplotlib, Jupyter Notebook
Embedded Systems: Linux (Ubuntu), ROS2, Jetson Orin Nano, Teensy, Arduino, UART, USB Serial, PWM, I2C, GPIO
AI/Developer Tools: Claude Code, GitHub Copilot, ChatGPT, Google Gemini
Certificates: Google Data Analytics, Oracle Cloud (OCI) Foundations, AWS Certified Developer - Associate (In Progress)

Projects

Chess Game Application | *Java, Spring Boot, PostgreSQL, Docker, AWS EC2, JUnit 5* Jul. 2024 – Present

- Led an initial **3-member** Java OOP project, then independently expanded it into a **client-server application** with a Swing GUI, Gradle multi-module build, and **Spring Boot REST API**.
- Built a **PostgreSQL Elo leaderboard** with game persistence and API-key-protected writes.
- Integrated a custom **negamax chess AI** with alpha-beta pruning and quiescence search for move evaluation and selection.
- Developed and maintained **20 JUnit 5 tests** using Mockito and Testcontainers; automated testing with **GitHub Actions CI**, orchestrated services with Docker Compose, and deployed the API to **AWS EC2**.

IEEE TXST Website Platform | *TypeScript, React, Next.js, Tailwind CSS, Vercel* Dec. 2025 – May 2026

- Led a **7-member** development team to replace a decade-old WordPress site with a production **Next.js 15** and **TypeScript** platform serving **100+ monthly users** and centralizing chapter events and resources.
- Developed **interactive event listings** with category filters, Google Forms RSVPs, and an embedded Google Calendar.
- Architected **reusable React components** and configuration-driven content for nontechnical maintainers; established **protected PR workflows**, automated Vercel deployments, and SEO optimizations.

MARS Robot | *Python, C++, ROS2, Jetson Orin Nano, Teensy 4.1, YOLOv11* Sep. 2025 – May 2026

- Directed a **17-member** IEEE RAS engineering team through Agile/Scrum to develop Python ROS2 software for perception, human tracking, and teleoperation; deployed a YOLOv11 Nano pipeline for real-time industrial object detection.
- Developed **C++ PWM motor-control firmware** on Teensy 4.1 and integrated it with NVIDIA Jetson-based **Python ROS2 software** through serial communication; tested and debugged hardware/software interfaces.

Experience

Geek Squad - Advanced Repair Technician Jun. 2026 – Present
Best Buy *San Jose, CA*

- Diagnose and resolve hardware/software failures across **PCs, Apple devices, and Linux systems**; document **Apple GSX** repair workflows, build computers, and communicate technical solutions as an **Apple-certified technician**.

IEEE & Boko Hackathon Leadership Aug. 2024 – May 2026
IEEE Student Branch, IEEE-HKN, IEEE RAS, Boko Hackathon *San Marcos, TX*

- Developed **C++ mecanum-control software** using PWM and encoder feedback, alongside **Python AprilTag computer vision**, on a 5-member team that placed **2nd out of 25 universities** at the IEEE Region 5 Robotics Competition.
- Served as IEEE Student Branch Vice President and IEEE-HKN Treasurer; co-founded IEEE Robotics & Automation Society (RAS), growing the chapter to **100+ members**.
- Co-directed an AI and cybersecurity hackathon with **100+ participants**, served as a technical code judge, and secured **$12,000+** in sponsorships from Google, Dell, Charles Schwab, IEEE, UFCU, and other organizations.

Team Lead and Learning Ambassador (Operations Associate) Jan. 2021 – Jul. 2025
Amazon *San Antonio, TX and Austin, TX*

- Led and trained a **10+ member** outbound operations team, monitoring supply-chain throughput of **10,000+ packages daily** and implementing process improvements that reduced truck dispatch time by **50%**.